

Project Report

by Varsha Shaji

Submission date: 08-Apr-2025 12:49AM (UTC+0530)

Submission ID: 2637964265

File name: FreelanceHub_report-plagiarism_check2.docx (5.92M)

Word count: 7169

Character count: 44257

ABSTRACT

FreelanceHub is a customized web-based freelance job managing system focused on enhancing collaboration between clients and freelancers. It allows posting and managing projects to be completed in Scrum framework. Additional, chat to be had in real-time, smart suggestions capabilities, conversation through AI technologies, and real time skill deficiency analysis with respective educational recommendations functions are accessible.

FreelanceHub enhances the way freelancers interact with the clients by giving them the opportunity to create rich profiles, join preconfigured teams with designated roles, and work together on projects. Clients can organize and advertise contests, issue diploma, and employ freelancers as workers. Trust is built by a compliment and rating system along with accountability. Payment through installments is made easier and secure.

FreelanceHub provides enhanced administrative features, enabling contest organization and user management while facilitating the resolution of disputes between clients and admins. Enhanced access is granted to those willing to subscribe to advanced features. Eliminate inefficiencies by blending thoughtful design, smart tools and collaborative workflows around the user. Secure Freelance working with clients professionally and engage in an enjoyable experience.

CHAPTER 1

INTRODUCTION

1.1 PROJECT OVERVIEW

FreelanceHub serves as a brand-new online marketplace that offers freelancers and clients the opportunity to improve the freelancing experience. Freelancers can search jobs, offer proposals, and work together while clients can post projects, review proposals, assign tasks, and just track progress. It is built on Scrum principles which means managing the workflow and team work is streamlined and everyone has defined roles like, Project Manager, Developer and Designer. It has key features including real time chat, AI chat bot support, smart project matching, skill gap analysis, event hosting, quizzes and issuance of certifications. FreelanceHub guarantees a reliable and scalable solution of modern project management and freelancer hiring with secure payments, transparent feedbacks and robust administrative tools.

1.2 PROJECT SPECIFICATION

FreelanceHub is web based freelance project management platform that allows clients to connect and perform work for freelance professionals in such a way to be efficient, secure and collaborative. It allows posting of projects, managing proposals, managing tasks, team collaboration, with AI support. Project workflows are based on the Scrum principles of transparency and productivity. Django (backend), MariaDB (database), HTML, CSS, and JavaScript are used as the frontend technologies and build the system.

Users in the Project

In Freelance Hub, there are three primary users:

- **Client:** Individuals or organizations posting job opportunities and managing projects.
- **Freelancer:** Independent professionals offering services and delivering project outcomes.
- **Administrator:** Oversees platform operations, manages user interactions, and ensures platform integrity.

Modules

Client Features

- **Profile Management** – Manage personal details and view past project history.
- **Project Posting** – Post detailed project requirements, scope, deadline, and budget.
- **Proposal Management** – View, accept, or reject freelancer proposals.
- **Task Management** – Define and manage tasks or user stories for projects.
- **Progress Tracking** – Monitor freelancer updates and project milestones.
- **Scrum Meeting Reminders** – Set and receive reminders for Scrum meetings.
- **Notifications** – Receive real-time alerts for updates and communication.

- **Chat System** – Communicate directly with freelancers through integrated chat.
- **Installment-Based Payment Processing** – Make secure milestone-based payments to freelancers.
- **Rating and Review** – Give feedback on freelancer performance after project completion.
- **Complaint Reporting** – Report misconduct or issues with freelancers.
- **Event Hosting** – Create and manage platform-wide events and quizzes.
- **Freelancer Hiring** – Search for and hire freelancers based on project needs.
- **Project Recommendations** – Get recommended freelancers matching project requirements.
- **AI Chatbot for FAQs** – Use AI support for quick answers to platform-related queries.
- **Premium Subscription Access** – Unlock additional features after posting three projects.
- **Password Reset** – Securely reset forgotten passwords.
- **Logout** – Safely log out of the platform.

Freelancer Features

- **Profile Management** – Set up and update skills, experience, and portfolio.
- **Project Search & Proposal Submission** – Search for projects and send personalized proposals.
- **To-Do List** – Track personal tasks and project-related deadlines.
- **Daily Progress Updates** – Provide clients with daily work updates.
- **Team Collaboration** – Form or join teams with assigned roles for collaborative work.
- **Rating and Review Clients** – Evaluate and review client interactions.
- **Complaint Reporting** – Report problems or issues with clients.
- **Notifications** – Receive instant alerts on updates and communication.
- **Chat System** – Communicate in real-time with clients via messaging.
- **Project Recommendations** – Get suggested projects based on skills and history.
- **AI Chatbot for FAQs** – Get automated answers to common platform queries.
- **Premium Subscription Access** – Unlock extra features after completing three projects.
- **Password Reset** – Recover access through secure password reset.
- **Logout** – Exit the account securely.

Administrator Features

- **User Management** – Oversee and manage all platform users and permissions.
- **Complaint Handling** – Investigate and resolve client-freelancer disputes.
- **Analytics Dashboard** – View insights on platform usage and performance.
- **Approve/Reject Events and Quizzes** – Moderate and approve hosted events and quizzes.

CHAPTER 2

SYSTEM STUDY

2.1 INTRODUCTION

A system analysis of FreelanceHub evaluates its platform architecture together with the user interaction method. It seeks to uncover how perfectly the system addresses the requirements of clients, freelancers and administratives with the help of project posting and proposal managing, along with collaboration. This study deals with the challenges, goals indication and system analysis methods to perform smoothly. The findings will enable the creation of a reliable and end user relevant freelance job management solution.

2.2 EXISTING SYSTEM

Currently, most of the freelance platforms offer basic functionalities such as posting a project, submitting a proposal and a messaging system. But, these tools and features are not integrated project management tools, real time collaboration features, and personalized recommendations. Communication gaps, lack of effective task tracking, and in efficient denouncing of disputes also negatively affect the user's experience. And most of the platforms also do not allow access to the advanced features until it's not clear what is the progression path for the users. On top of that, no team collaboration, no AI support, not a lot of structured workflows such as Scrum are usually lacking or very poorly developed. These limitations depict that there is a lack of a comprehensive system to do the above.

NATURAL SYSTEM STUDIED

Within the natural system, clients and freelancers might meet in the informal networks, on social media or simply through personal referral. Payment and communications are usually through email or messaging apps, and payments are still done manually. Due to task tracking, performance monitoring and accountability being limited, instead of delays, miscommunication, and lack of transparency.

DESIGNED SYSTEM STUDIED

On the other hand, there are some existing platforms meant for freelancers like Upwork, Fiverr, and Freelancer that offer just rudimentary services — anything from projects posting and proposal submission; that messaging. However, these are the systems which provide the opportunity of online freelancing, but mostly these systems do not provide a good advanced project management, AI support, real time task tracking etc., facilities. There are many locked premium tiers that have limited free access to many of their essential tools.

2.2.1 DRAWBACKS OF EXISTING SYSTEM

- Limited support for Scrum-based project management and task progress tracking.
- No structured team collaboration or defined roles of any project.
- Absence of intelligent skill gap analysis or personalized project recommendations.
- Lack of AI support for frequent queries that are asked or for system guidance.

2.3 PROPOSED SYSTEM

FreelanceHub is a proposal of full featured web based freelance job management platform for clients and freelance managers to avoid the inessential appointment process. The project execution here is done on Scrum principles, communication is done through real time chat, AI powered chat bots are used for query resolution and finally this uses the installment based payment methods. The system features team collaboration modules, hosting of events and quizzes, evaluation of skill gaps, and project recommendations in accordance with user profiles. Users are also shown other administration tools such as managing users, resolving complaints, and maintaining platform integrity.

2.3.1 ADVANTAGES OF PROPOSED SYSTEM

- A project management based on scrum to manage structured and effective task execution.
- Chat, daily progress updates etc. are real time collaboration tools.
- Team formation with defined roles like Project Manager, Developer, QA, etc.
- AI enabled chatbot to assist users on getting quick answers to FAQs.
- Recommendations of smart project and freelancer based on account data.
- Secure, installment-based payment processing with transparency.
- Event and quiz management with certificate issuance and approval system.

CHAPTER 3 REQUIREMENT ANALYSIS

3.1 FEASIBILITY STUDY

Information system feasibility determines if the proposed design and implementation of the solution can be both accomplished successfully. A feasibility study determining the practicability of the solution involves factors such as resources, cost and benefits that are likely to be incurred. This is mainly because there are several common classes of feasibility that are often taken into consideration when developing a management information system. Following are the main ones.

3.1.1 Economical Feasibility

The assessment of economic viability checks whether software development expenses and running expenses can be offset by its benefits. Yes, Freelance Hub can be considered to be economically feasible because of no license and development costs due to use of open- source technologies. New revenue sources will include; subscription services where freelancers and clients will be required to pay some amount after they post three jobs to ensure they pay for more services in any given fiscal year. Since it does not require a lot of money to start and as the company clearly understands how it can generate revenue, it could be highly likely to become highly profitable in the long-run.

3.1.2 Technical Feasibility

Technical viability looks at whether the present technology, resources, and structures of the firm are capable of meeting project specification before the set point of time or budget. Freelance Hub platform is indeed quite viable when it comes to its technological approach, using Django on the back end, while the front end is coded in HTML, CSS and JavaScript. Consequently, MariaDB 5 provides clients with high online transactional processing and capable of data scalability. There are objective opportunities for developing integrated technologies that can enable future growth at the same time as integrated efficient development.

3.1.3 Behavioral Feasibility

Operational feasibility refers with the implementation of the developed system in the real-world environment within the set business objectives and user specifications using currently available manpower resources in the organization. Organizationally, Freelance Hub is plausible to operate; this is because it enhances communication between the clients and freelancers. Real time communicate and the in-app notifications, help the users in keeping them updated and in touch. Event moderation is customizable and can be undertaken by the administrators of the said platform. All these features collectively make it possible for the platform to operate, improve efficiency, and be easily usable after implementation.

3.1.4 Feasibility Study Questionnaire

Project Overview:

The Freelance Hub will resolve the problems related to managing the freelance jobs by introducing a complete web-based solution. It tackles problems like slow job posting, poor communication between client and freelancer, and management of payment and project progress. These improvements are aimed at optimizing user satisfaction, streamlining operations for clients and freelancers, and promoting collaboration within the freelance ecosystem.

System Scope:

It is suggested that The Freelance Hub will be a production-ready implementation. It will include all the basic features needed to effectively manage freelance jobs and will be scalable and reliable enough to handle a growing user base and growing overall transaction volumes over time.

Target Audience:

Users of the Freelance Hub generally consist of:

- **Clients:** Users that are looking for freelancers in all kinds of industries.
- **Freelancers:** Self-employed specialists providing their services and knowledge.
- **Administrators:** Manage user accounts and disputes and maintain the integrity and security of the platform.

Modules:

Nuts and Bolts of the Freelance Hub include:

- **Project Posting and Management:** Freelancers bid on projects posted by clients and both parties manage their project listings and proposals.
- **Proposals and Project Management:** Client post a project and receive freelancer recommendations, while freelancers submit proposals
- **User Profile Management :** Clients and freelancers create and manage their profiles, where premium features are unlocked after posting/completing projects.
- **Real-time Chat System:** The clients and freelancers discuss the projects through chat, video, and audio calls.
- **Payment Processing:** Secure payment transactions are processed, with updates and reminders sent for client and freelancers.
- **Rating and Review System:** Freelancers are rated by clients for performance which aids in increasing trust and accountability.

- **Admin Dashboard:** Create and Manage user accounts, Monitor User Activity, Fraud Detection, Approve/Reject events.
- **FAQs AI ChatBot:** Both clients and freelancers can get instant replies to any queries on the platform here, thanks to an AI chatbot.
- **Events Creation:** customers can create events and quizzes then give prizes.

User Roles:

FreelanceHub user roles are defined as:

- **Client:** Can create projects, manage proposals, and communicate with freelancers, see the work progress, release payment, and give feedback.
- **Freelancer:** Search for projects, submit proposals, communicate with clients, manage tasks and milestones, receive pay, and build their reputation through ratings and reviews from clients.
- **Administrator:** Manages the user account, oversees the operation of the whole platform, deals with disputes, ensures compliance with platform policies, optimizes for user experience.

System Ownership:

The Freelance Hub is developed by an academic institution for a project purpose.

Industry/Domain:

The Freelance Hub works in the freelancing job management sector, serving multiple industries that require freelance services, such as technology, creative arts, and others.

Data Collection Contacts:

Name: Linoy Stephen

Role: Director

Contact information: 9633200034

Company: Nanobird Technologies

Questionnaire for Data Collection:**1. What is the most popular freelance jobs that are being requested at this time?**

The four main freelance job categories consist of web development alongside graphic design and content writing and digital marketing.

2. How do clients normally decide which freelancer should work for their projects?

This means clients usually choose a freelancer relying on resume, related experiences, portfolio, reviews and rating from added clients as well as proposal.

3. When the freelancers search for freelance job opportunities what do they usually look for?

They are freelancers who try to search for jobs they would be eligible for using their skills, experience, pay they would be getting, and clear project descriptions. At the same time, they enjoy clients who have a history of payments on time and positive feedback of other freelancers.

4. What is the preferred way freelancers communicate to clients throughout the work process?

The easiest and the best way to communicate for freelance people is by using integrated real time chat system inside the platform. Additionally, they use email and video calls for broader discussions and project updates.

5. How is it usually done, payment between clients and freelancers?

Usually, the payments are taken care of through secure payment gateways like Stripe or PayPal. The payment by clients mostly involves paying a percent upfront and the rest is paid at the completion of the project.

6. How do you think freelancers find most useful features or tools available on freelance job platforms in your experience?

The most useful features include specific job postings, effective proposal management, instant chat system, secure handling of payment, as well as its rating and review system for freelancers.

7. Is the evaluation of clients and freelancers decisive in the field of freelance work?

Feedbacks and ratings are significant since they increase the level of reliability. It would be noted that freelance workers with high rankings enjoy better prospects of being hired because most clients are more inclined to consult services that possess good ratings because ratings act as indicators to the efficiency and credibility of freelancers.

8. It is crucial to know some vital characteristics that may enable the freelance job platform to support users in finding the most suitable employers and effective job searches.

Some of the necessities are as follows: a search system, profile management system, security measures for payment gateway, a rating and review system, real-time communication methods, among others.

9. How do clients and freelancers usually resolve disputes or issues?

Disputes are typically resolved through the platform's resolution center, where both parties can

present their case. Administrators or mediators then review the information and make a decision.

10. Which of the following features should be included for communication in the platform, that contains chatting, e-mail, and video conferencing?

However, real-time chat has the potential to bring this convenience of communication, you may further incorporate an email for giving more formal notification or video call option for getting a clear vision of the projects, in a discussion with the clientele or even the freelancers.

11. When it comes to the dissemination of the updates and milestones in the project, how should the freelancers be communicating this to the clients?

One of the issues involves realizing timely project updates and milestones. We recommend:

- Focused Corporate/Project Specific Instrumentation: It is possible to introduce the concepts of Corporate/Project-specific dashboard/boards to track the progress, calendar, and timelines.
- Reports: Automatic creation of progress reports and reports with progress bar and accomplishments.
- Notification: This involves making notification within the application regarding important events within the system, deadlines, completed activities among others.

12. Which of these methods of user verification are appropriate (email, phone, face verification)?

Phone OTP and email verification is enough to ensure adequate security, but it is recommended to add an ID check as well. Additional security is possible by incorporating face verification in the function.

13. Which premium features to include (e.g., boosting profile, unlimited proposals)?

Scalable features provide motivation for users, and premium features provide additional capabilities. Consider these options:

- Profile Expansion: Provide extra profile visibility to attract clients to freelancers.
- Unlimited Proposals: Give freelancers the ability to submit unlimited proposals to boost their chances of securing work.
- Priority Support: Discerning support channels with a lesser hold up time for premium customers.
- Individual Advanced Analytics: Provide in-depth analytics on user activity and project insights.

14. Will the platform take a cut for each job posting, transaction, or membership?

For a platform to be sustainable, it needs to generate revenue. Consider the following fee structures:

- Project Posting Fees — Charge clients fees to post jobs.
- Commission Fees: Take a tiny percentage on successful transactions between projects.
- Membership Fees: Introduce tiered membership plans offering different levels of benefits and features.

15. What notifications do we want users to receive (project updates, payment reminders, etc.)

Notification should cover: Project updates and Milestones, Payment reminders and successful payments, Proposal acceptance or rejection New job postings based on preferences.

16. What notification types (email or in-app or WhatsApp, etc.) should use?

Distribute notifications through several mediums as per user preference:

- In-app Notifications: For real time and eye-catching notifications.
- Email Notifications: More formal full length update.

17. Will the platform be a site for hosting events like webinars, quizzes, or networking sessions to prompt engagement?

Yes, hosting webinars, workshops, virtual meet-ups, and events to engage users to provide value-added services.

18. How would you manage team collaboration for a project on the platform and which roles would you include (manager, developer, designer etc.)? What particular duties should each role be responsible for as well?

When dealing with complex projects, a crucial aspect is its collaboration features. Consider the following:

- Project Management Tools: Link project management tools like Trello or Asana to set out tasks, assign responsibilities, and determine progress.
- Team Roles: Specify roles such as Project Manager, Team Lead, Developer, Designer, Quality Assurance for clear responsibilities.
- Assign and Track Tasks: Empower project managers to allocate tasks to team members, monitor progress, and give feedback.
- Communication channels: Prevent noise with dedicated channels for each team and/or project.

19. Is it possible for clients to hire freelancers directly on the platform?

Yes, the clients will also have the option to hire freelancers directly from their profile or via the proposals they submit for the job postings.

20. Should AI chatbots help users with common questions and platform navigation?

Yes, AI chatbots can help users find answers for commonly asked questions, help users explore the platform, and even provide technical assistance.

21. What new technologies (e.g., AI, machine learning) could enhance platform capabilities?

- **AI-driven Skill Matching:** Include AI capable of matching freelancers to client projects in a better dimension due to experience, skills, and project requirements.
- **Predictive Analytics:** Use machine learning for predicting project timelines; analyzing potential risks; suggesting optimal project workflows;
- **Fraud Detection:** Use AI algorithms to identify fraudulent activities, such as fake profiles or disputes over payments.
- **Personalized Recommendations:** Use AI to suggest the most appropriate freelancers, projects, and training materials to users.

22. Do you have any unique features you want to incorporate that are not usually found on like platforms?

- **Skill Assessment Tests** — Implement skill assessment tests to validate freelancer skills and offer unbiased performance evaluations.
- **Portfolio Showcase:** Allow freelancers to present their work in an attractive and interactive way.
- **Client Reviews and Testimonials:** Develop a strong mechanism to allow clients to review and leave testimonials for freelancers, to foster confidence and credibility.
- **Mentorship Program** — Create a mentorship program that actually connects experienced freelancers with aspiring ones.

3.1.5 Geotagged Photograph



3.1 SYSTEM SPECIFICATION

3.2.1 Hardware Specification

Processor - Intel core i5

RAM - 8GB

Hard Disk - 500GB SSD

3.2.2 Software Specification

Front End - HTML, Bootstrap, JavaScript, jQuery, AJAX, CSS

Back End - Django

Database - MariaDB

Client on PC - Windows 7 and above

Technologies used - HTML, JavaScript, CSS, Django, MariaDB, AJAX, NLP, Google Gemini API, Email API, Razor pay API, Dynamic Portfolio Creation from resume, Course Recommendations based on skill gap

3.3 SOFTWARE DESCRIPTION

3.3.1 Django

Django functions as a Python web framework to create secure and scalable web applications at a high rate of speed. Django uses a Model-View-Template (MVT) architectural design and incorporates default capabilities for security authentication, database control, and URL routing, which enhance coding speed. Django implements a reusable design that reduces duplicate code and follows the implementation standard of "Don't Repeat Yourself" (DRY).

3.3.2 MariaDB

The database system MariaDB functions as an open-source relational database management system (RDBMS) which started as a split from MySQL. It's known for its reliability, performance, and scalability, which makes it a well-liked option for web applications. MariaDB supports SQL for database queries and offers additional features, improved security, and performance optimizations compared to MySQL. It is fully compatible with MySQL, allowing seamless migration between the two. MariaDB is commonly used with frameworks like Django to manage and store large volumes of structured data.

1 CHAPTER 4 SYSTEM DESIGN

4.1 INTRODUCTION

In system design, a deep outline of a system's architecture, components, and data flow along with other elements is created to fulfill prescribed functional and non-functional requirements. It deals with its performance, scalability, security, usability, and maintainability. A system's design should be such that it works with optimal efficiency and can be modified easily in the future. The interaction diagrams alongside system structures get represented through different diagrams including use case, class, sequence and activity diagrams. Along with these diagrams, ER diagrams aid in data visualization, and together they help data developers better communicate their design during the design stage.

4.2 UML DIAGRAM

UML diagrams are typical visual representations of interactions and activities within or within a system's organization. In UML diagrams, classes, objects, and relations may be presented with certain shapes like circles and rectangles. Developers, software analysts and other stakeholders can use these diagrams for documenting, building, designing and even reasoning about softwares. These diagrams are versatile and assist in almost every activity during the software development lifecycle. This adaptive quality guarantees the meeting of user needs and requirements during system development. From simple applications to complex enterprise systems, everything can be modeled using UML diagrams.

4.2.1 USE CASE DIAGRAM

A use-case diagram demonstrates system usage by different user types which entails their accessible system functions according to their assigned roles. It defines system boundaries which help emphasize essential features and enables early user requirement acquisition throughout development stages to guarantee system goal achievement.

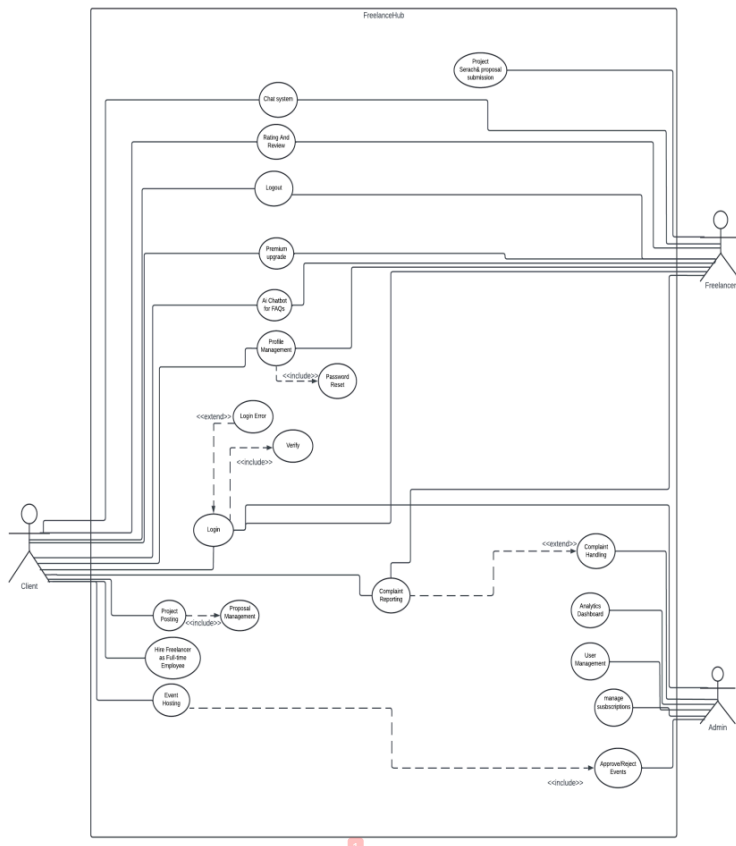


Figure 1: Use case diagram

4.2.2 SEQUENCE DIAGRAM

The diagram illustrates how objects and components communicate with each other throughout a particular scenario. Vertical lifelines represent objects on the diagram and horizontal arrows symbolize the message transfers between these objects. The display of control flow enables developers to improve software design along with system interaction analysis and produce development documentation.



Figure 2: Sequence diagram

4.2.3 State Chart Diagram

In UML, **State Chart Diagram** or the **State Machine Diagram** represents an object along with its multiple states and trigger conditions these states use to change. Through State Chart Diagrams we can identify how systems operate dynamically because the diagrams track objects through their operational states. Rounded rectangles define the states and arrows between these states illustrate state transitions and events determine shifts between states and the diagram includes initial and terminating states. Real-time systems and embedded systems use state chart diagrams for model developments in addition to their use in workflow modeling.

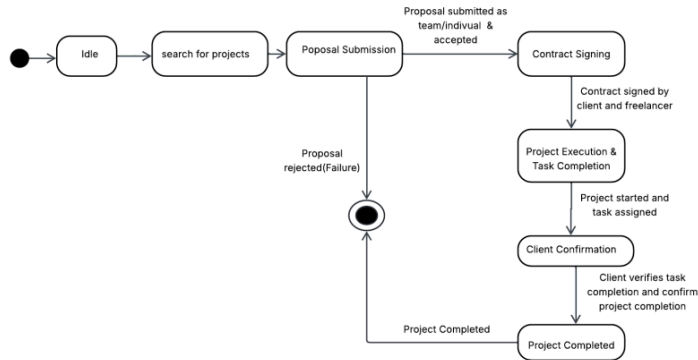


Figure 3: State chart diagram

4.2.4 Activity Diagram

Activity diagrams enable visual workflow modeling through which data and control flow can be represented to explain process behavior. The diagram displays system and use case activities as well as decisions along with their execution order. Subsequent to programming and business process transformation activity diagrams represent complex workflows which allow stakeholders to understand system behavior and processes better.

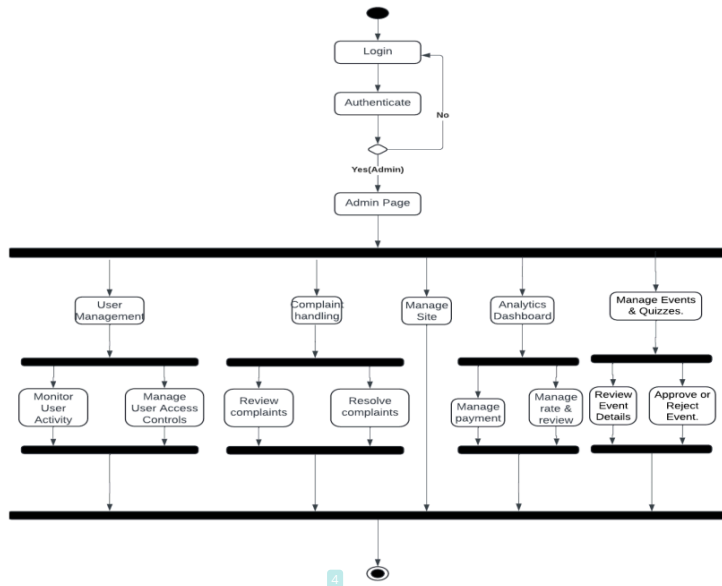


Figure 4: Activity diagram of admin

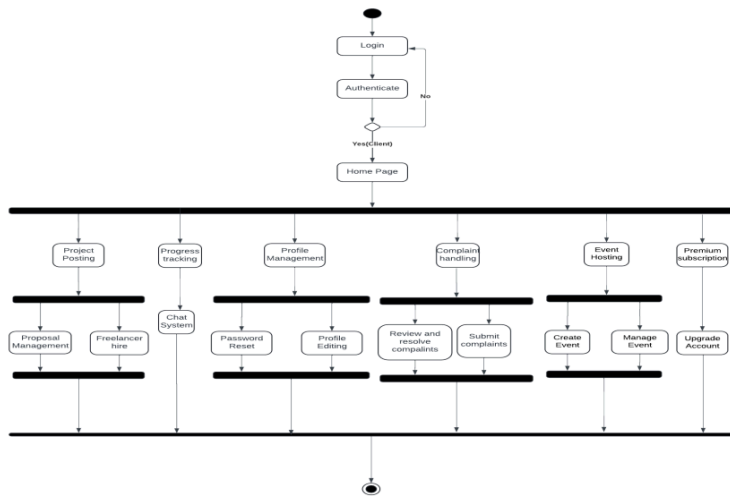


Figure 5: Activity diagram of client

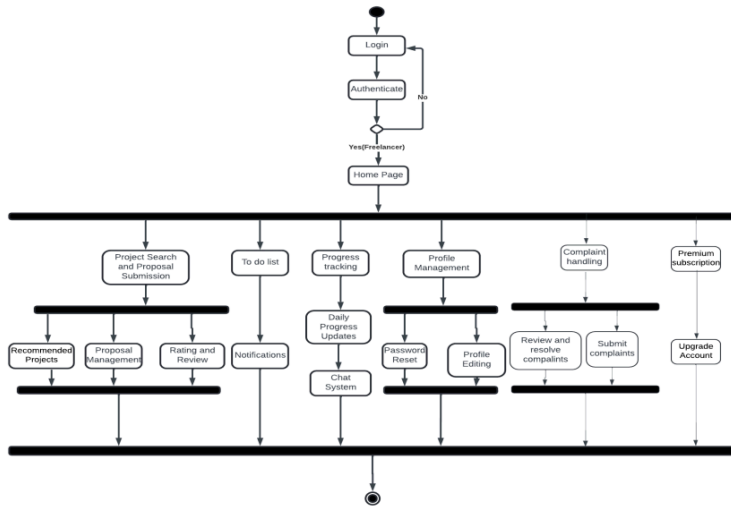


Figure 6: Activity diagram of freelancer

4.2.5 Class Diagram

The object-oriented programming framework uses class diagrams to represent system structures with their classes alongside their attributes and methods and relationships. The diagram helps developers create system architecture designs while describing class interaction patterns and defining essential properties with their associated behaviors. The development process relies on class diagrams as essential tools which maintain developer-stakeholder-team collaboration from planning to communication throughout project development.



Figure 7: Class diagram

4.2.6 Object Diagram

During runtime execution the modeling technique offers instant views of system instances joined by their associations at one single point in time. The diagram depicts real-world classes through their objects together with their attribute values as well as object-to-object relationship links. The behavior of a system during execution becomes visible through object diagrams so developers can utilize them for validating class structures and real-time interactions while understanding and analyzing the system.



Figure 8: Object diagram

4.2.7 Component Diagram

The diagram displays the system structure through an illustration of both software components and their connection points and system dependencies. The diagram enables engineers to understand system interaction patterns between modules and APIs and external systems thus supporting the development of modular maintainable software designs. Each component becomes a rectangular shape with connecting interfaces that shows developers how various parts interact and organize within the system design.

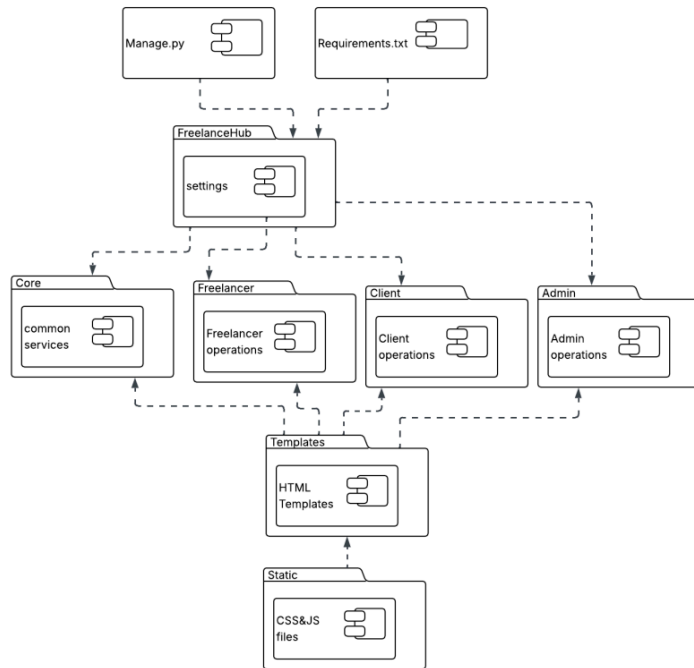


Figure 9: Component diagram

4.2.8 Deployment Diagram

A physical architecture of the system, how the system takes software components to hardware nodes (servers, devices or cloud environment) is represented in a Deployment Diagram using UML. The relationships between the nodes (hardware), artifacts (software components), and communication paths are depicted through it. It aids in system architects as well as DevOps teams' understanding of the system's infrastructure, network distribution, and the deployment strategy.

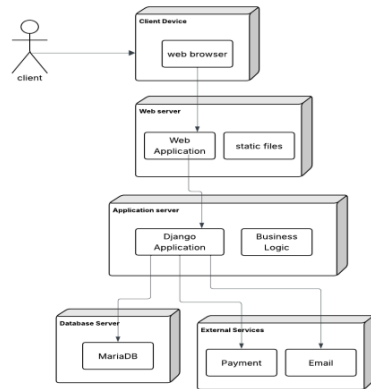


Figure 10: Deployment diagram

4.3 USER INTERFACE DESIGN USING FIGMA

Form Name: Login

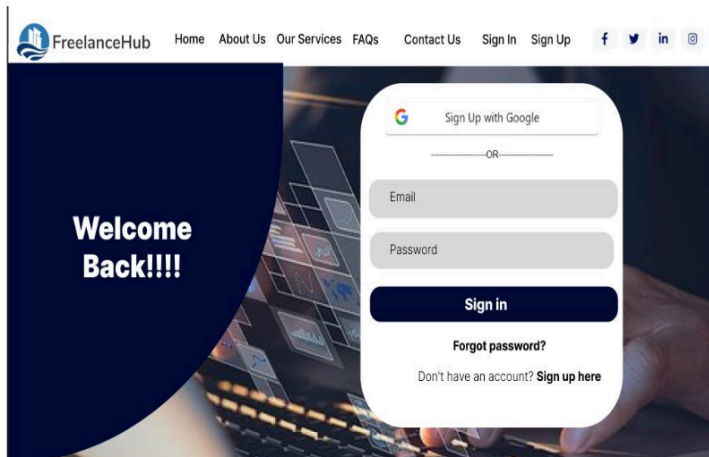
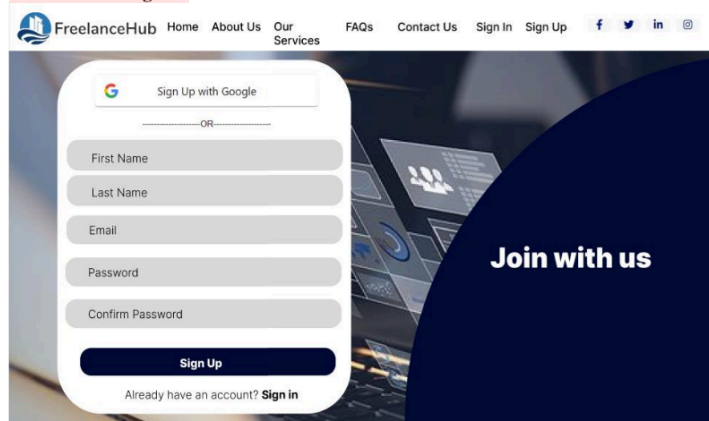
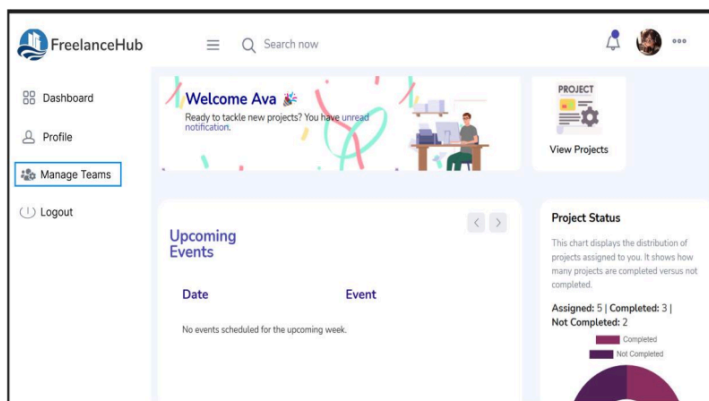


Figure 11: Login page

Form Name: Register

The screenshot shows the 'Register' page of the FreelanceHub website. The page features a dark blue background with a laptop and a 'Join with us' text. A white registration form is overlaid on the left. The form includes a 'Sign Up with Google' button, an 'OR' separator, and input fields for 'First Name', 'Last Name', 'Email', 'Password', and 'Confirm Password'. A 'Sign Up' button is at the bottom of the form. Below the button, it says 'Already have an account? Sign in'. The top navigation bar includes links for Home, About Us, Our Services, FAQs, Contact Us, Sign in, and Sign Up, along with social media icons for Facebook, Twitter, LinkedIn, and Instagram.

Figure 12: Register Page

Form Name: Freelancer Dashboard

The screenshot shows the 'Freelancer Dashboard' of the FreelanceHub website. The dashboard has a light blue header with the FreelanceHub logo, a search bar, and user profile icons. A left sidebar contains navigation links: Dashboard, Profile, Manage Teams (highlighted), and Logout. The main content area includes a 'Welcome Ava' message with a notification, a 'PROJECT' section with a 'View Projects' button, an 'Upcoming Events' section with a table header (Date, Event) and a message 'No events scheduled for the upcoming week.', and a 'Project Status' section with a pie chart and text: 'Assigned: 5 | Completed: 3 | Not Completed: 2'. The pie chart legend shows 'Completed' in purple and 'Not Completed' in dark purple.

Figure 13: Freelancer's Dashboard

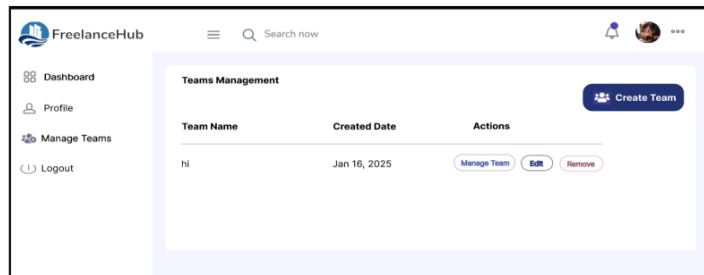
Form Name: All Team management Page

Figure 14: Teams management page

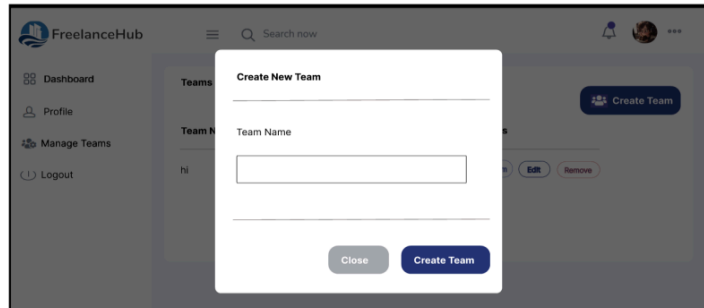
Form Name: New team creation

Figure 15: New Team Creation form

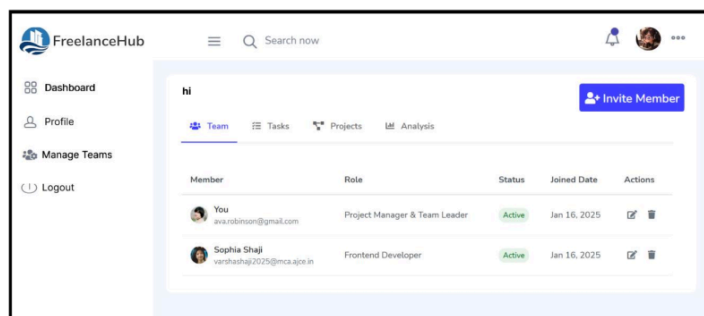
Form Name: Team management Page

Figure 16: Individual team management page

Form Name: Member Invitation Form

Figure 17: Team member invitation form

4.4 DATABASE DESIGN

4.4.1 Relational Database Management System (RDBMS)

A relational system manages structures data by connecting fixed tables through predefined relationships among them. Tables in an RDBMS maintain distinct fields for data attributes which the SQL language manipulates to execute insertions and updates and make deletions and query operations. Data consistency and integrity in RDBMS function through constraints. Worldwide organizations use this database system to handle structured data volumes efficiently because it provides scalable resources, dependable functionality and methodical data organization.

4.4.2 Normalization

A database organization process named Normalization functions to remove unnecessary data repetitions while enhancing system data quality. Large tables become smaller manageable tables through this approach which also includes defining the relationships between these tables. The database normalization process contains several stages with unique standards which function to eliminate repeated data storage and maintain data unity throughout one storage point. These measures minimize data inconsistencies and anomalies thus supplying better efficiency and accuracy for database operations.

Regularization divides into multiple stages or "normal forms" that define organizational rules for data structures. The most common normal forms are:

- First Normal Form (1NF)
- Second Normal Form (2NF)

- Third Normal Form (3NF)
- Boyce-Codd Normal Form
- Fourth Normal Form
- Fifth normal form

First Normal Form (1NF)

First Normal Form (1NF) exists as the base form of database normalization to enforce a unique primary key and atomic (indivisible) value in all table columns. The First Normal Form ensures data separation into individual tables which eliminates repeated groups while improving database integrity and reducing administrative complexity.

For example:

The STUDENT relation in table 1 violates 1NF illustrating STUD_PHONE as its multi-valued attribute. The decomposition process leading to 1NF for the STUDENT table is presented in table 2.

STUD_NO	STUD_NAME	STUD_PHONE	STUD_STATE	STUD_COUNTRY
1	RAM	9716271721, 9871717178	HARYANA	INDIA
2	RAM	9898297281	PUNJAB	INDIA
3	SURESH		PUNJAB	INDIA

Table 1

Conversion to first normal form

STUD_NO	STUD_NAME	STUD_PHONE	STUD_STATE	STUD_COUNTRY
1	RAM	9716271721	HARYANA	
1	RAM	9871717178	HARYANA	INDIA
2	RAM	9898297281	PUNJAB	INDIA
3	SURESH		PUNJAB	INDIA

Table 2

Second Normal Form (2NF)

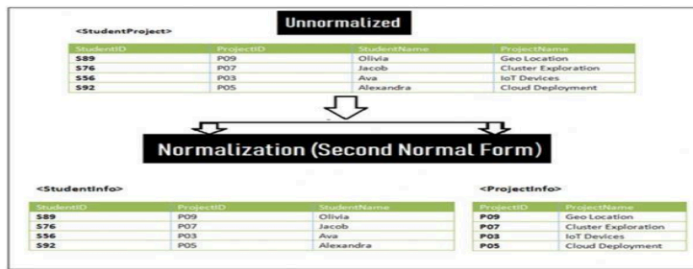
The Second Normal Form (2NF) requires complete dependence of all non-key attributes on full primary keys to minimize table dependencies when multiple attributes make up the primary key. The data maintains higher integrity while it reduces redundancy because the organization method supports easier query execution and simpler maintenance operations.

For Example:

StudentID and ProjectID represent the primary key attributes of the relation.

The non-key attributes StudentName along with ProjectName require functional dependence on a candidate key portion to qualify as Partial Dependent.

Each StudentName can be identified through StudentID allowing the relation to qualify as Partial Dependent. ProjectID serves as a means to determine the ProjectName in this relation thus making it Partial Dependent. The database design for <StudentProject> contains issues according to 2NF rules in Normalization which makes this design unsatisfactory.

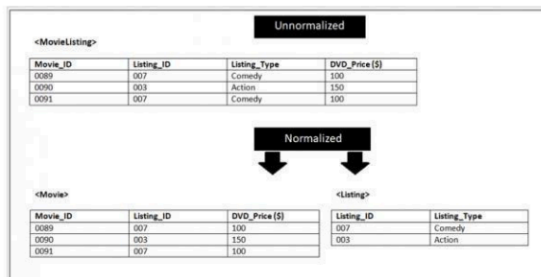


Third Normal Form (3NF)

The objective of Third Normal Form (3NF) is to restrict all non-key attributes in a table from depending on anything but the primary key to prevent transitive dependencies. Achieving these data management principles reduces unneeded data duplications and strengthens the overall data reliability. Table splitting occurs as a common operational step when attempting to reach 3NF standards. The database structure may become complex due to this enhancement yet the use case determines if normal form implementation is necessary.

For Example:

The table maintains a violation of Third Normal Form (3NF) because it includes a transitive dependency where Movie_ID causes Listing_ID which leads to Listing_Type. The identification of Movie_ID shows an indirect relationship with Listing_Type since it establishes transitive functional dependencies.



Boyce-Codd Normal Form

A database requires every determinant under the BCNF rule to function as a candidate key. Functional dependencies between non-key attributes that are not trivial must be removed to decrease database anomalies and prevent data inconsistencies. The achievement of BCNF standard demands that tables are split which adds complication but delivers superior data consistency while

cutting down duplicate storage. The complete benefits become apparent with this method when it is utilized in extensive database systems since it optimizes query execution and minimizes potential data errors.

The scenario uses relation $R(A, B, C)$ to illustrate this case.

$$\begin{aligned} A &\rightarrow B \\ A &\rightarrow C \\ B &\rightarrow A \end{aligned}$$

Because keys A and B represent super keys we can confirm that the above mentioned relation fulfills BCNF standards.

Fourth Normal Form

Fourth Normal Form (4NF) advances **Third Normal Form (3NF)** through its approach to solve multi-valued dependencies inside database structures requiring **non-key attributes** to functionally **depend on the primary key** as the **only** criterion while removing all non-trivial multi-valued dependencies. A multi-valued dependency exists when different independent non-key attributes separate from each other by a primary key.

Table breakup during this procedure eradicates repeated data and upholds database consistency while preventing data inconsistency.

A typical case where 4NF application applies to tables is in "Student Course Schedule" information tables. The table named "Course_Schedule" contains three primary key combinations with the following fields:

- Student_ID (PK)
- Course_ID (PK)
- Semester
- Grade

This table enables multiple course enrollments for each student at once along with supporting one course being taught to different students at same time. The relationship between students and courses consists of multiple students taking numerous courses at once. The instructor teaching a course becomes different throughout each semester while the courses themselves show variation based on instructor assignments within a semester.

The relationship between Course_ID and Instructor fields exists in multiple values because of this design. Normalization of the table to 4NF requires splitting it into the two separate entities: Course_Schedule and Course_Instructor.

1. Course_Schedule table:

- Student_ID (PK)

- Course_ID (PK)
- Semester
- Grade

2. Course_Instructor table:

- Course_ID (PK)
- Semester (PK)
- Instructor

Our normalization brought a solution to the multivalued dependency issue while making every entry in our database single-valued without duplication. The independent updates of the Course_Instructor table will maintain dataset consistency with the Course_Schedule table contents.

Fifth normal form

The Project-Join Normal Form which tackles complicated many-to-many connections breaks down tables to delete superfluous data and keeps data consistency. Join operations allow reconstructing tables which leads to better consistency and flexibility. The attainment of 5NF delivers benefits yet its establishment involves substantial complexity together with possible impractical aspects.

4.4.3 Sanitization

Database security depends on sanitization through the process which cleans problematic inputs thus defending databases from SQL injection and XSS attacks. Database security together with integrity maintenance becomes possible through input validation and parameterized queries and prepared statements. Database systems require proper sanitization methods to create secure dependability.

4.4.4 Indexing

In database structure design indexing creates data organization systems which use column values for quick retrieval by identifying the database locations. The system assists users in finding necessary data rapidly without needing to review entire databases which boosts query execution speeds notably in big or intricately designed databases. Selective usage of indexes remains crucial because multiple indexes negatively affect write operations speed and expand storage needs.

CHAPTER 5

SYSTEM TESTING

5.1 INTRODUCTION

Quality Assurance identifies bugs as well as deviations within the application early and making sure that software is of the highest quality, reliability and functionality. Unit, integration, system, acceptance and regression testing are the key types. It enhances the security, performance and quality standards compliance. Testing minimizes risks and lowers costs by identifying problems before deployment and helps improve the customer satisfaction which results into a robust and reliable product.

5.2 TEST PLAN

A testing strategy is an official outline which defines the approach of testing an application. It specifies test environment, test cases, tools, data and team roles in order to guide the testing team. So, the main goal is the process of testing structured and efficiently. In addition, it contains informative details on the type of testing to be done at various levels, helping you scan, locate and eliminate problems systematically. This usually involves:

- Unit Testing
- Integration Testing
- System Testing
- Acceptance Testing

5.2.1 Unit Testing

Technique used to assess software components that tests code components or function in isolation. It guarantees that the individual module performs correctly according to its requirements. Automated test cases are a tool used by developers to catch issues early in the development process and thereby code easier to maintain, reliable and easier to debug.

5.2.2 Integration Testing

Integration testing tests how various parts of a system combine together in regard to the flow of data between modules and how they interact with each other. Built on top of the unit testing, it helps to find issues that aren't caught by unit tests, guarantees a clean integration of subsystems and external interfaces. Top down, bottom up or even hybrid methods are the common approaches. The intention is to verify that when combined parts are used together that the system behaves properly before it leaves the lab.

5.2.3 Validation Testing or System Testing

The validation testing is intended to test the software to confirm if it works as per stakeholder's and user's needs. While system testing assesses all of the entire system's functionality, reliability,

and interaction amongst components. Together they are a key part on making sure the software is complete, functional, and ready to be released.

5.2.4 Output Testing or User Acceptance Testing

It is the final phase in the testing and the statements will be made by the end users to ensure that software meets the business needs and is performing as expected. It guarantees that the system is operation-friendly, and reliable with functional as well as non-functional requirements. That is to say UAT of the software confirms that it is ready for deployment.

5.2.5 Automation Testing

Automation testing is done using tools, scripts to carry out automatically test cases so it is faster, accurate and consistent than the manual testing. This is useful for functional and regression testing, especially it gives early detection ability to identify problems, reduction of costs, and concomitantly improves overall software quality.

5.2.6 Selenium Testing

Selenium functions as a freely accessible automation toolset to control web browser interactions and it serves as a common solution for functional and regression testing of web applications. It is able to support multiple languages of programming including Java, C#, and Python, and it can simulate user interactions like typing and clicking on various browsers such as Chrome and Firefox. With features of creating test scripts, running and reporting Selenium is a well-integrated with an open source existing TestNG, Junit, Jenkins and others to aptly suit the environment of the software and its industry.

Example:

Test Case 1

Code

Screenshot

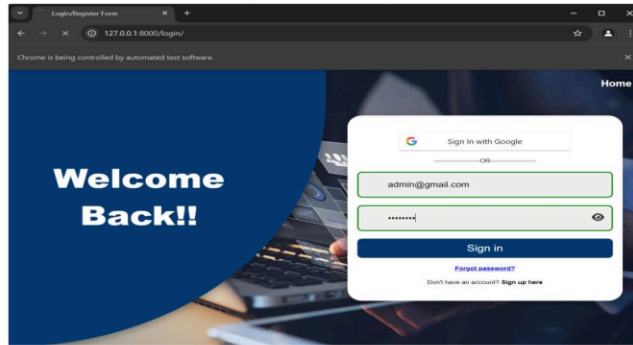


Figure 18: Login page having credentials entered

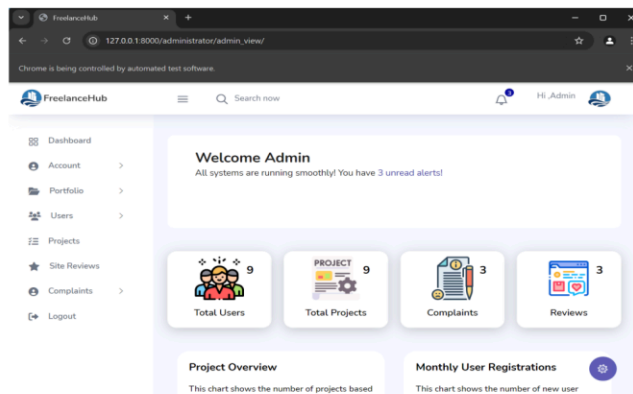


Figure 19: Admin dashboard page after successful Login

```
PS C:\Users\LENOVO\Desktop\freelancehub> & "C:/Program Files/python/python.exe" c:/Users/LENOVO/Desktop/freelancehub/tests/test_login.py
DevTools listening on ws://127.0.0.1:58105/devtools/browser/5b40227f-9736-4eae-8947-e6d39eb8d896
Login successful for admin@gmail.com.
Created TensorFlow Lite XNNPACK delegate for CPU.
Attempting to use a delegate that only supports static-sized tensors with a graph that has dynamic-sized tensors (tensor#58 is a dynamic-sized tensor).
Login successful for ava.robinson@gmail.com.
Login successful for samuel@gmail.com.
Login failed. Incorrect URL after login.
```

Figure 20: Test result of login attempt

Test Case 2:**Code****Screenshot**

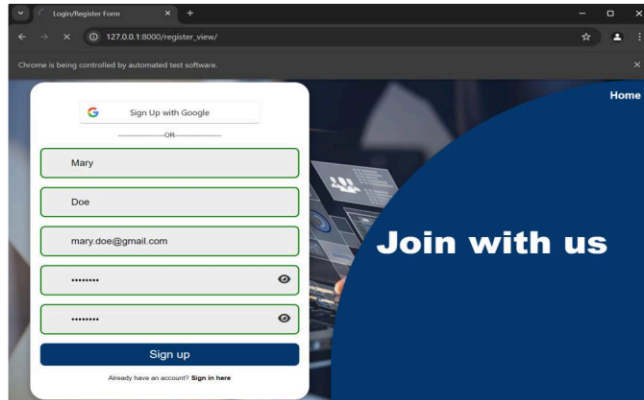


Figure 21: Register page with valid credentials entered

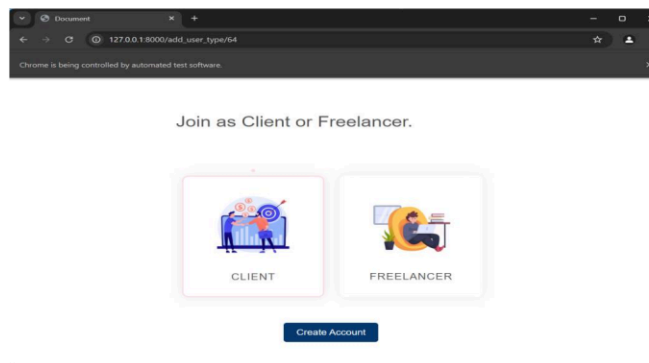


Figure 22: Role choose page after successful registration

```
PS C:\Users\LENOVO\Desktop\freelancehub> & "C:/Program Files/python/python.exe" c:/Users/LENOVO/Desktop/freelancehub/tests/test_register.py  
DevTools listening on ws://127.0.0.1:58097/devtools/browser/0250af10-83d5-4bb3-8ff6-26a6c817f703  
Test Passed :Registration successful for mary.doe@gmail.com  
Created TensorFlow Lite XNNPACK delegate for CPU.  
Attempting to use a delegate that only supports static-sized tensors with a graph that has dynamic-sized tensors (tensor#58 is a dynamic-sized tensor).  
Test passed for existing email noah.wilson@gmail.com
```

Figure 23: Register page Test result

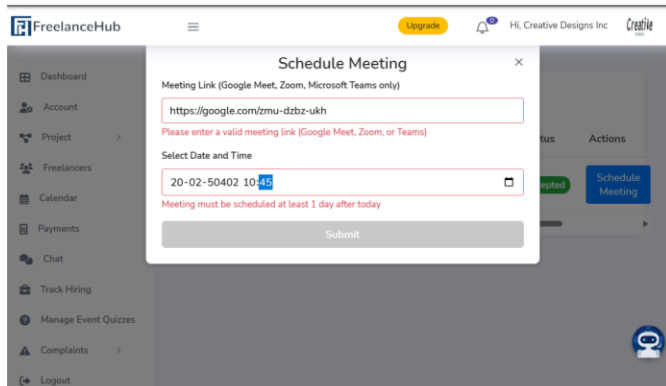
Test Case 3:**Code****Screenshot**

Figure 24: Meeting schedule form with invalid data

```
PS C:\Users\LENOVO\OneDrive\Desktop\freelancehub> & "C:/Program Files/python/python.exe" c:/Users/LENOVO/OneDrive/Desktop/freelancehub/tests/test_meeting_scheduling.py

DevTools listening on ws://127.0.0.1:53354/devtools/browser/043994b5-f7c7-40db-9f74-bafc45d526d8
✔ Test Passed: Submit button is disabled with an invalid meeting link.
```

Figure 25: Meeting schedule form Test result

Test Case 4:**Code****Screenshot**

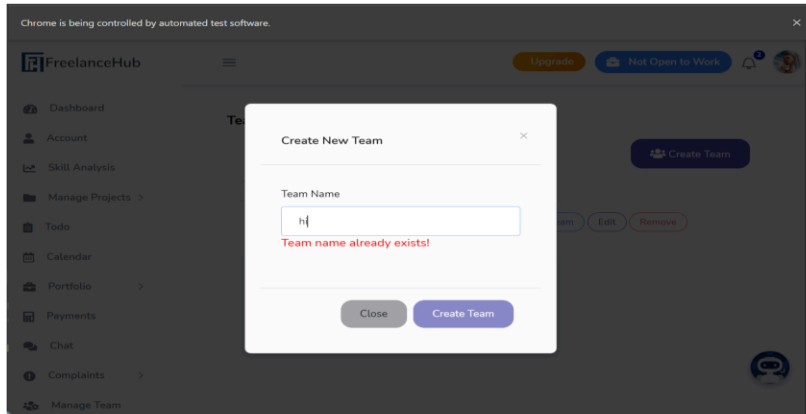


Figure 26: Team creation form

```
PS C:\Users\LENOVO\OneDrive\Desktop\Freelancehub> & "C:/Program Files/python/python.exe" c:/Users/LENOVO/OneDrive/Desktop/freelancehub/tests/test_team_forms.py
fed-a343-5c413f425e03
Logged in successfully.
Test Passed: Submit button is disabled for existing team name.
Created TensorFlow Lite XNNPACK delegate for CPU.
Attempting to use a delegate that only supports static-sized tensors with a graph that has dynamic-sized tensors (tensor#1 is a dynamic-sized tensor).
Test Passed: Team created successfully with new name.
```

Figure 27: Team creation form Test result

Test Case 5:**Code****Screenshot**

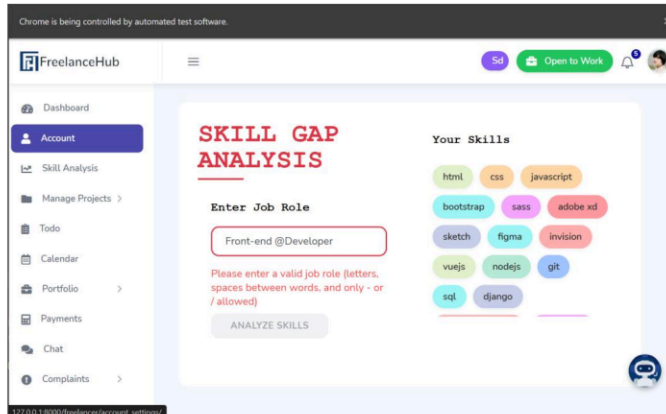


Figure 28: Skill gap analysis form with invalid data entered

```
PS C:\Users\LENOVO\OneDrive\Desktop\freelancehub> & "C:/Program Files/python/python.exe" c:/Users/LENOVO/OneDrive/Desktop/freelancehub/tests/test_skill_gap.py  
DevTools listening on ws://127.0.0.1:62982/devtools/browser/cb29a2ec-731a-40c1-929d-781235ad5aa2  
Test Passed: Button disabled and error shown for invalid input 'Front-end @Developer'
```

Figure 29: Skill gap analysis form Test result

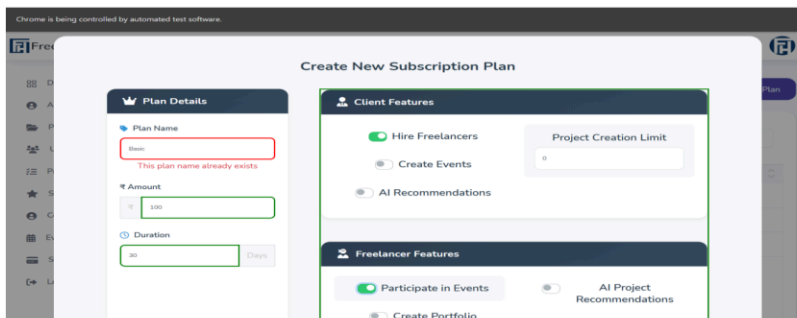
Test Case 6:**Code****Screenshot**

Figure 30: subscription plan creation form with invalid data

```
PS C:\Users\LENOVO\OneDrive\Desktop\freelancehub> & "C:/Program Files/p
python/python.exe" c:/Users/LENOVO/OneDrive/Desktop/freelancehub/tests/t
est_subscription_plan.py

DevTools listening on ws://127.0.0.1:54659/devtools/browser/4e70f923-0f
ab-4cc9-9b37-a5b76a9eaa68
✓ Logged in successfully.

Testing Form Validation:
✓ Test Passed: Create Plan button is enabled with valid data.
```

Figure 31: subscription plan creation form Test result

CHAPTER 6

IMPLEMENTATION

6.1 INTRODUCTION

Software design becomes a functional product by developing it through programming and testing operations within the same phase. During this phase developers both verify requirements fulfillment and apply strategies to maintain the software along with making it scalable. The target platform receives the product after stakeholders cooperate for proper planning. A system that meets these requirements becomes a stable and functional product which remains easy to maintain at its current state.

6.2 IMPLEMENTATION PROCEDURES

Software implementation begins with planning, goal setting, task assignment, and setup. Key steps include:

- **Development & Testing:** Developers build and test the software for functionality.
- **User Acceptance Testing:** Real users confirm it meets their needs.
- **Deployment:** Final version is released to users.
- **Post-Deployment Testing:** System is tested in live conditions.
- **Support & Maintenance:** Issues are fixed and improvements are made.
- **Feedback & Improvement:** User input guides future enhancements.
- **Collaboration:** Ongoing communication ensures smooth execution.

6.2.1 User Training

Effective software implementation depends heavily on user training because it allows end-users to handle the system effectively. There are four standard procedures for user training:

- **Training Needs Assessment:** Identify which software features users need to learn.
- **Training Plan Development:** Create a plan with goals, methods, and materials.
- **Training Delivery:** Conduct training via classes, online modules, or one-on-one sessions.
- **Training Evaluation:** Gather user feedback to measure effectiveness.
- **Follow-Up Training:** Offer additional sessions to address remaining skill gaps.

User training programs empower end-users to operate new software effectively thus raising productivity while enabling smooth business transitions. The implementation process requires enough training and support to reach successful adoption and generates better business performance.

6.2.2 Training on the Application Software

The effectiveness and efficiency of end-users depend on receiving training for application software

applications. The key steps include:

- The first step is orientation through which staff members receive a basic overview about software capabilities together with its advantages to spark their enthusiasm.
- Through practical training opportunities users gain experience by conducting real sessions or testing procedures within the software framework.
- Training sessions adapt to user job roles in order to keep content relevant to their work responsibilities.
- The training program includes user-specific materials that provide easy-to-follow user-friendly information.
- Monitoring training success and improving effectiveness occurs during the evaluation step.

User confidence increases together with workplace productivity while the software implementation transition becomes easier through effective training methods.

6.2.3 System Maintenance

The system maintenance function ensures both operational efficiency and business requirements fulfillment within a project. Key activities include:

- **Regular Updates** – Apply bug fixes, security patches, and new features to maintain efficiency.
- **Backup and Recovery** – Perform regular data backups and have a disaster recovery plan.
- **Performance Monitoring** – Track system performance to detect and resolve issues.
- **User Support** – Provide help desk assistance for user queries and technical issues.
- **Security Measures** – Implement firewalls, antivirus, and intrusion detection for protection.
- **Training and Documentation** – Offer ongoing training and update user manuals as needed.

The correct maintenance practices lead to better operational efficiency together with enhanced protection and uninterrupted business operations.

6.2.4 Hosting

Koyeb provides reliable and scalable access to the Django-based FreelanceHub project. The combination of MariaDB database service through Files.io supports automatic scaling and deployment features alongside secure storage functions. The system operates with strong protection and excellent performance through SSL encryption and database backup processes along with monitoring capabilities.

Procedure for hosting on Koyeb.app:

- Create an account at Koyeb and log in.

- Click "Create Service" and connect your GitHub repository.
- Ensure Django runs locally and create requirements.txt using pip freeze > requirements.txt.
- Create a Procfile with web: `gunicorn myproject.wsgi --bind 0.0.0.0:$PORT` (replace myproject).
- Set ALLOWED_HOSTS in settings.py and configure the database credentials for MariaDB.
- Configure static files using WhiteNoise or external storage.
- Click Deploy on Koyeb and monitor logs for issues.
- Access the live app via the Koyeb-provided URL.
- Push updates to GitHub for automatic redeployment.

Hosted Website: Koyeb.app, render.com

Hosted Link:

- <https://freelancehub.koyeb.app/>
- <https://freelancehub-5tya.onrender.com/>

Hosted Link QR Code



Screenshot

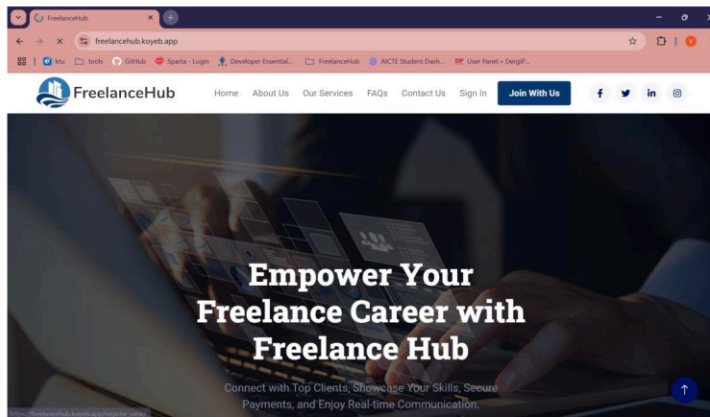


Figure 32: Hosted website

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

FreelanceHub serves as a complex Freelance Job Management System which simplifies team operations to boost communication between clients and freelancers. Both project posting services and proposal management and agile execution with Scrum framework run on the system platform. The system enables real-time chatting as it supports AI device assistance during skill assessment procedures to create individual training programs. The platform allows freelancers to create credentials that lead to joining collaborative team projects and clients acquire capabilities to plan events and quiz activities while managing employee recruitment with certification capability. Transaction security comes from the payment feedback system together with installment payments that ensure financial protection for transactions. FreelanceHub provides strong administrative options with improved premium features that establish a protected and organized interactive platform for freelance work.

7.2 FUTURE SCOPE

This includes AI-based job matching that it says could allow freelancers to discover relevant opportunities faster. Contract generation automation will form contracts faster than ever between clients and freelancers. Improved team collaboration tools will allow multi-user projects management without a hitch. Interactive learning programs, skill badges, and premium job visibility will also be available on the platform. By harnessing advanced analytics, intelligent hiring tools, and a developing AI ecosystem, FreelanceHub will deem to take a thriving, effective, and creative freelance industry to new heights.

Project Report

ORIGINALITY REPORT

12%	6%	1%	10%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Amal Jyothi College of Engineering Student Paper	10%
2	Submitted to ESoft Metro Campus, Sri Lanka Student Paper	<1%
3	www.coursehero.com Internet Source	<1%
4	Submitted to University of Wolverhampton Student Paper	<1%
5	daw.dbmanagement.info Internet Source	<1%
6	edshare.gcu.ac.uk Internet Source	<1%
7	lbms03.cityu.edu.hk Internet Source	<1%
8	www.analyticsvidhya.com Internet Source	<1%
9	Jon Stephens, Chad Russell. "Chapter 3 Keys, Indexes, and Normalization", Springer Science and Business Media LLC, 2004 Publication	<1%
10	dev.to Internet Source	<1%
11	Bui Thanh Hung, M. Sekar, Ayhan Esi, R. Senthil Kumar. "Applications of Mathematics	<1%

12	Submitted to Colorado Technical University Online Student Paper	<1 %
13	docer.tips Internet Source	<1 %
14	epdf.pub Internet Source	<1 %

Exclude quotes	Off	Exclude matches	Off
Exclude bibliography	On		